13/11/2024

Phase-2 Code-B

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 | NCERT Booster Test Series for NEET (RM)-2024-25_T02B | Time : 200 Min.

Topics Covered:
Physics: Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves, Electric Charges & Fields
Chemsitry: Thermodynamics, Equilibrium, Redox Reactions, Electrochemistry, Solutions
Botany: Morphology of Flowering Plants, Anatomy of Flowering Plants, Plant Kingdom
Zoology: Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration, Animal Kingdom-I: Non-chordates (Upto Aschelminthes)

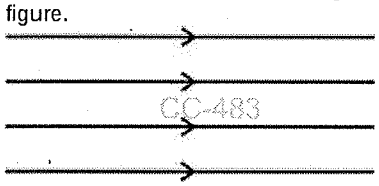
General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. Electric field lines in a certain region of space are shown in figure.



Then, choose the correct option.

(1) Both magnitude and direction of electric field are uniform

(2) Both magnitude and direction of electric field are non-uniform

(3) Only magnitude of electric field is uniform

(4) Only direction of electric field is uniform

2. **Statement I:** Total charge of an isolated system remains conserved.
Statement II: Charge is a scalar quantity.

(1) Both statements I and II are true and statement II is correct explanation of statement I

(2) Both statements I and II are true but statement II is not correct explanation of I

(3) Statement I is true and statement II is false

(4) Both Statements I and II are false

3. If 10^{16} electrons are removed from a neutral metal sphere, then the charge on the sphere will be

(1) +1.6 mC

(2) –1.6 mC

(3) +3.2 mC

(4) –3.2 mC

4. If an electric field is given by $\left(5\hat{i}+4\hat{j}+3\hat{k}\right)$, then the electric flux through a surface of area 6 units lying in *xy* plane will be

(1) 18 units

(2) 36 units

(3) 24 units

(4) 30 units

5. The unit of quantity $\frac{g}{G}$ in SI unit will be (symbols have their usual meaning)

(1) $kg^2\ m^{-1}$

(2) $kg^{-1}\ m^2$

(3) $kg\ m^{-2}$

(4) $kg^{-2}\ m$

6. A planet has a mass six times the mass of earth. If *g* is acceleration due to gravity on earth's surface, then acceleration due to gravity on planet surface is (Radius of planet and earth is same)

(1) 6*g*

(2) 4*g*

(3) 8*g*

(4) 16*g*

7. The time period of an earth satellite in circular orbit is independent of

(1) Radius of orbit

(2) Mass of satellite

(3) Mass of earth

(4) All of these

8. A body of mass 100 g experiences a gravitational force of 4 N, when placed at a particular point. The magnitude of the gravitational field intensity at that point is

(1) 0.04 N/kg

(2) 400 N/kg

(3) 40 N/kg

(4) 25 N/kg

9. Modulus of elasticity depends on

(1) Nature of material

(2) Shape and size of material

(3) Force applied on the material

(4) Both (2) & (3)

10. A cable that can support a maximum load of 240 kg is cut into two equal parts. The maximum load that can be supported by either part is

(1) 60 kg

(2) 120 kg

(3) 240 kg

(4) 480 kg

11. The elastic energy stored in a wire of Young's modulus *Y* is

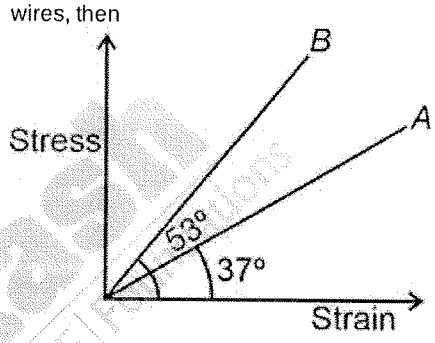
(1) Stress × Strain × Volume

(2) $\frac{\text{Stress}^2 \times \text{Volume}}{2Y}$

(3) $\frac{1}{2} \times Y \times$ Volume

(4) $Y \times \frac{\text{Strain}^2}{\text{Volume}}$

12. The stress-strain graphs for wires *A* and *B* are shown in figure. If Y_A and Y_B are Young's modulus of the material of wires, then



(1) $Y_A = 2Y_B$

(2) $16Y_A = 9Y_B$

(3) $9Y_A = 16Y_B$

(4) $Y_A = Y_B$

13. The cylindrical tube of a spray pump has a cross -section of 8 cm^2. One end of tube has 40 fine holes each of area 10^{-8} m^2. If the liquid flows inside the tube with a speed of 0.15 m/min, then find the speed with which the liquid is ejected through the holes.

(1) 20 m/s

(2) 5 m/s

(3) 10 m/s

(4) 2 m/s

14. Bernoulli's theorem is based on the principle of

(1) Conservation of mass

(2) Conservation of energy

(3) Conservation of angular momentum

(4) Conservation of linear momentum

15. Excess pressure inside a soap bubble of radius r is proportional to

(1) r

(2) r^0

(3) r^{-1}

(4) $r^{-1/2}$

16. The amount of work done to form a soap bubble of radius R is [S = surface tension of soap bubble]

(1) $8\pi R^2 S$

(2) $\frac{3}{2}\pi R^2 S$

(3) $24\pi R^2 S$

(4) $4\pi R^2 S$

If the length of a cylinder on heating increases by 2%, the area of its base will increase by

(1) 0.5%

(2) 2%

(3) 1%

(4) 4%

18. 100 g ice at 0°C is mixed with 100 g water at 80°C. The final temperature of the mixture is

(1) 40°C

(2) 25°C

(3) 0°C

(4) 47.14°C

19. A faulty thermometer reads the temperature of melting ice and boiling water as 40°F and 200°F respectively. What will be its reading when actual temperature is 41°F?

(1) 43°F

(2) 40°F

(3) 42°F

(4) 48°F

20. The coefficient of thermal conductivity (K) depends on

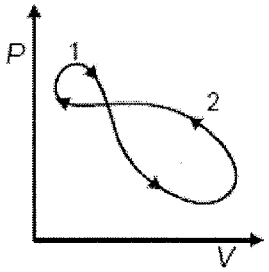
(1) Thickness of the sample

(2) Material of the plate

(3) Temperature of the sample

(4) Area of the sample

21. In the following indicator diagram, the net amount of work done by the gas will be (P : pressure, V : volume)



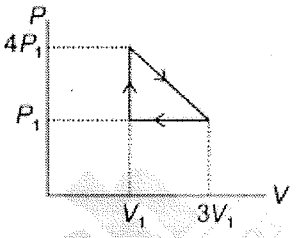
(1) Positive

(2) Infinity

(3) Zero

(4) Negative

22. The amount of work done in the following cyclic process is



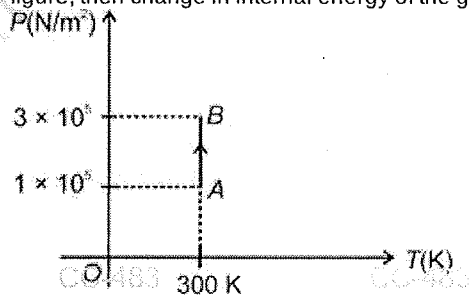
(1) P_1V_1

(2) $3P_1V_1$

(3) $6P_1V_1$

(4) $12P_1V_1$

23. If 2 mole of an ideal gas is taken from A to B as shown in figure, then change in internal energy of the gas is



(1) –2376 J

(2) +2376 J

(3) –1188 J

(4) Zero

24. $\Delta Q = \Delta U$ is valid for

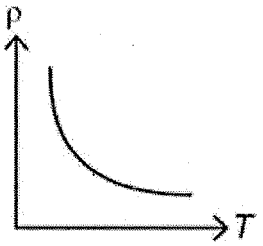
(1) Isothermal process

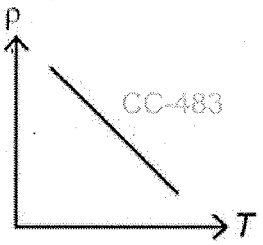
(2) Adiabatic process

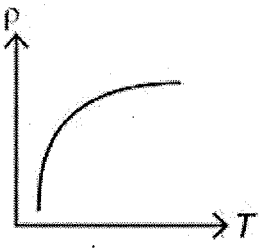
(3) Isochoric process

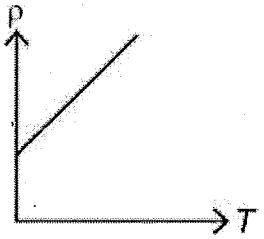
(4) Isobaric process

25. Which variation among the following shows the correct relationship between the temperature (T) and density (ρ) of an ideal gas assuming pressure to be constant?

(1) 

(2) 

(3) 

(4) 

26. The average translational kinetic energy of a gas molecule is

(1) Proportional to pressure of gas

(2) Inversely proportional to volume of gas

(3) Inversely proportional to absolute temperature of gas

(4) Proportional to absolute temperature of gas

27. The translational degrees of freedom of an ideal diatomic gas is

(1) 2

(2) 3

(3) 5

(4) 4

28. A cylindrical tube, open at both ends, has a fundamental frequency f in air. If $\frac{1}{3}$ of the length is dipped vertically in water, the fundamental frequency of the air column will be

(1) $\frac{3f}{2}$

(2) $\frac{3f}{4}$

(3) f

(4) $\frac{f}{2}$

29. A particle is performing simple harmonic motion given by equation $y = 4\sin\omega t + 3\cos\omega t$, then the amplitude of the SHM will be

(1) 2 units

(2) 7 units

(3) 1 unit

(4) 5 units

30. A particle executes SHM of amplitude 10 m and period 4 s along straight line. The velocity of the particle at a distance 6 m from mean position is

(1) 2 π m/s

(2) 4 π m/s

(3) 6 π m/s

(4) 5.4 π m/s

31. The displacement of a particle x and time t are related as $\frac{d^2x}{dt^2} + C^2x = 0$. Time period of particle

(1) $2\pi C^2$

(2) $\frac{2\pi}{C^2}$

(3) $\frac{2\pi}{C}$

(4) $2\pi C$

32. In a stationary wave produced in a string, all the particles

(1) On either side of a node vibrate in the same phase.

(2) In the region between two consecutive nodes vibrates in the same phase.

(3) In the region between two antinodes vibrate in the same phase.

(4) Of the medium vibrate in the same phase.

33. The fundamental frequency of a closed organ pipe is same as the 1st overtone of an open pipe. If the length of open pipe is 50 cm, then the length of closed pipe is

(1) 25 cm

(2) 12.5 cm

(3) 100 cm

(4) 200 cm

34. The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is

(1) 140 cm

(2) 80 cm

(3) 100 cm

(4) 120 cm

35. The equation of a progressive wave is $y = 25 \sin(20t + 5x)$, where y is displacement, hence the incorrect statement is that the

(1) Amplitude of the wave is 25 units

(2) Wave is propagating in positive x-direction

(3) Velocity of wave is 4 units

(4) Maximum velocity of particle is 500 units

SECTION-B

36. Electric field at a distance R from infinite uniformly charged sheet is proportional to

(1) R^0

(2) $R^{\frac{3}{2}}$

(3) $R^{\frac{1}{2}}$

(4) R^2

37. A particle of mass m and charge q is placed at rest in a uniform electric field E and then released. The kinetic energy attained by the particle after moving a distance y is

(1) qEy

(2) qEy^2

(3) qE^2y

(4) q^2Ey

38. If A is the areal velocity of a planet of mass m, its angular momentum is

(1) $2mA$

(2) A^2m

(3) Am^2

(4) $\frac{m}{A}$

39. If two wires of same material and same length have ratio of their radii 1 : 2, then under the effect of same stretching force their respective elongation will be in ratio

(1) 1 : 1

(2) 1 : 2

(3) 4 : 1

(4) 1 : 8

40. A 20 cm long capillary tube is dipped in water, the water rises upto 8 cm. Now this arrangement is placed in a lift, which is accelerating upward with acceleration g. The height of water column in capillary tube is

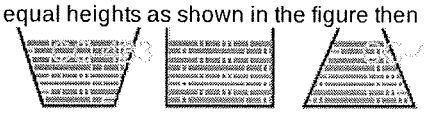
(1) 8 cm

(2) 4 cm

(3) 6 cm

(4) 2 cm

41. A liquid is poured in the three vessels of same base area to equal heights as shown in the figure then



(1) Maximum force on the base will be for vessel C

(2) Maximum force on the base will be for vessel B

(3) Maximum force on the base will be for vessel A

(4) Force on the base will be equal for all the vessels

42. If a bimetallic strip is heated, it will

(1) Bend towards the metal with lower thermal expansion coefficient

(2) Bend towards the metal with higher thermal expansion coefficient

(3) Twist itself into helix

(4) Have no bending

43. The temperature of a body changes from 60°C to 40°C in 7 minutes and temperature of surroundings being 10°C. Its temperature after next 7 minutes is (Assuming Newton's law of cooling to be valid)

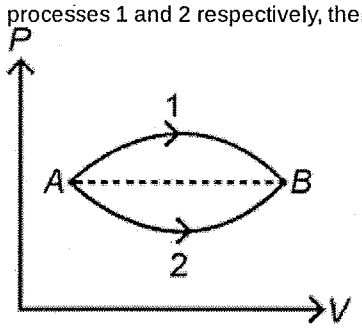
(1) 28°C

(2) 48°C

(3) 50°C

(4) 78°C

44. A system goes from *A* to *B* via two processes 1 and 2 as shown in the figure. If ΔQ_1 and ΔQ_2 are the heat supplied in processes 1 and 2 respectively, then



(1) $\Delta Q_1 > \Delta Q_2$

(2) $\Delta Q_1 = \Delta Q_2$

(3) $\Delta Q_1 < \Delta Q_2$

(4) $\Delta Q_1 \leq \Delta Q_2$

45. The concept of temperature is defined by

(1) First law of thermodynamics

(2) Second law of thermodynamics

(3) Third law of thermodynamics

(4) Zeroth law of thermodynamics

46. Ratio of mean translational kinetic energy of He and O_2 gas molecules at a temperature will be

(1) 3 : 5

(2) 1 : 2

(3) 1 : 1

(4) 4 : 5

47. A particle executes SHM with angular frequency $\omega = 4\pi$ rad/s. If it is at its extreme position initially, then the time period of oscillation of particle is

(1) 1.5 s

(2) 0.5 s

(3) 4 s

(4) 2.3 s

48. Equation of a particle executing simple harmonic motion is $y = 0.4 \sin\left(2\pi t + \frac{\pi}{3}\right)$ (where, t is in second and y is in metre). Initial phase of the particle is

(1) $2\pi t + \frac{\pi}{3}$

(2) $\frac{\pi}{3}$

(3) $\frac{7\pi}{3}$

(4) $\frac{8\pi}{3}$

49. The equation of simple harmonic progressive wave given by $y = 0.5\sin(2\pi t - 2\pi x)$, then equation of a wave with double the amplitude and half frequency but travelling in opposite direction in same medium will be

(1) $y = 1.0\sin(2\pi t + 2\pi x)$

(2) $y = 1.0\sin(\pi t + 2\pi x)$

(3) $y = 1.0\sin(\pi t + \pi x)$

(4) $y = 0.5\sin(\pi x - \pi t)$

50. Equation of a stationary wave is $y = 0.6 \sin(314t)\cos(3.14x)$, where x is in metre and t is in second. Distance between two consecutive nodes is

(1) 2 m

(2) 1.5 m

(3) 1 m

(4) 0.5 m

CHEMISTRY

SECTION-A

51. Intensive property among the following is

(1) Temperature

(2) Volume

(3) Enthalpy

(4) Gibbs free energy

52. A path function among the following is

(1) Temperature

(2) Internal energy

(3) Enthalpy

(4) Heat

53. A 5 litre cylinder contained 440 g of CO_2 gas at 27°C. After some time, it was found that the cylinder is empty as the whole gas has leaked out. If the atmospheric pressure is 1 atm then the work done by the gas is

(1) 5 L-atm

(2) -241.3 L-atm

(3) 0 L-atm

(4) 413.2 L-atm

54. For the reaction $A_4(s) \longrightarrow 4A(g)$ if $\Delta U = 2$ kcal and $\Delta S = 10$ cal K^{-1} at 27°C then the ΔG for the given reaction is [R = 2 cal mol^{-1} K^{-1}]

(1) 1 kcal

(2) 1.4 kcal

(3) 5.6 kcal

(4) 7.4 kcal

55. If the enthalpy of formation of $N_2O(g)$, $N_2O_4(g)$, $CO(g)$ and $CO_2(g)$ is 800, 15, –100, and –400 kJ/mol respectively, then Δ_rH for the given reaction is

$$N_2O_4(g) + 3\,CO(g) \rightarrow N_2O(g) + 3\,CO_2(g)$$

(1) +90 kJ/mol

(2) –115 kJ/mol

(3) +115 kJ/mol

(4) –500 kJ/mol

56. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A): Entropy of perfectly crystalline solid is zero at zero kelvin.
Reason (R): At 0°C all molecular motion ceases.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

(2) **(A)** is true but **(R)** is false

(3) **(A)** is false but **(R)** is true

(4) Both **(A)** and **(R)** are true and **(R)** is the correct explanation **(A)**

57. Which among the following match is incorrect?

	$\Delta_rH°$	$\Delta_rS°$	$\Delta_rG°$	**Description**
(1)	–	+	–	Reaction spontaneous at all temperature
(2)	–	–	–	Reaction spontaneous at low temperature
(3)	+	+	–	Reaction nonspontaneous at high temperature
(4)	+	–	+	Reaction nonspontaneous at all temperature

(1) (1)

(2) (2)

(3) (3)

(4) (4)

58. Consider the following statements.
Statement (I): All measurable properties of the system remain constant at equilibrium.
Statement (II): Equilibrium is possible only in a open system at a given temperature.
Choose the correct option

(1) Statement **(I)** is correct but Statement **(II)** is incorrect

(2) Both Statement **(I)** and Statement **(II)** are correct

(3) Statement **(I)** is incorrect but Statement **(II)** is correct

(4) Both statement **(I)** and statement **(II)** are incorrect

59. At 300 K, the value of $\left(\frac{K_c}{K_p}\right)$ will be lowest for

(1) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

(2) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(3) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(4) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

60. Consider the following equilibrium
$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$
If the partial pressure of ammonia at equilibrium is 4 atm and total pressure is 20 atm then the equilibrium constant (K_p) of the reaction will be

(1) 9 atm^2

(2) 81 atm^2

(3) 432 atm^2

(4) 32 atm^2

61. At a certain temperature and pressure of 4 atm, N_2O_4 is 20% dissociated. The pressure (in atm) at which N_2O_4 will be 40% dissociated will be
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(1) 0.123

(2) 0.875

(3) 0.441

(4) 0.835

62. Match Column-I with Column-II

Column-I (sparingly soluble salts)	Column-II Solubility product (K_{sp}) in terms of solubility (s)
(i) AB_2	(a) $108s^5$
(ii) A_2B_3	(b) $27s^4$
(iii) AB	(c) $4s^3$
(iv) AB_3	(d) s^2

Choose the correct option.

(1) (i)-(d), (ii)-(b), (iii)-(c), (iv)-(a)

(2) (i)-(c), (ii)-(a), (iii)-(d), (iv)-(b)

(3) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)

(4) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b)

63. Consider the following statements
Statement I : Phosphoric acid has more than one ionizable proton per molecule of acid.
Statement II : Higher order ionization constants (K_{a_2} and K_{a_3}) are higher than the lower ionization constant (K_{a_1}) for phosphoric acid.
In the light of above statements, choose the correct option.

(1) Statement **(I)** is correct but Statement **(II)** is incorrect

(2) Both Statement **(I)** and Statement **(II)** are correct

(3) Statement **(I)** is incorrect but Statement **(II)** is correct

(4) Both statement **(I)** and statement **(II)** are incorrect

64. At 25°C, pH of a 10^{-8} M aqueous KOH solution will be

(1) 7.04

(2) 6.02

(3) 8.02

(4) 9.02

65. The equilibrium constant K_C for the reaction is 4×10^2 at 300 K. The $\Delta G^{\ominus}$ of the reaction is

(1) + 6.48 kJ mol^{-1}

(2) + 14.96 kJ mol^{-1}

(3) – 6.48 kJ mol^{-1}

(4) – 14.96 kJ mol^{-1}

66. The minimum volume of water required to dissolve 4.08 gram of Calcium Sulphate at 298 K is [Given K_{sp} of Calcium Sulphate is 9×10^{-6}]

(1) 2.5 L

(2) 5.0 L

(3) 10.0 L

(4) 12 L

67. Match Column-I with Column-II.

Column-I	Column-II
(i) $2KClO_3(s) \xrightarrow{\Delta} 2KCl(s) + 3O_2(g)$	(a) Non-metal displacement redox reaction
(ii) $V_2O_5(s) + 5Ca(s) \xrightarrow{\Delta} 2V(s) + 5CaO(s)$	(b) Decomposition redox reaction
(iii) $Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$	(c) Combination redox reaction
(iv) $CH_4(g) + 2O_2(g) \xrightarrow{\Delta} CO_2(g) + 2H_2O(l)$	(d) Metal displacement redox reaction

Choose the correct option.

(1) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)

(2) (i)-(b), (ii)-(d), (iii)-(a), (iv)-(c)

(3) (i)-(b), (ii)-(a), (iii)-(d), (iv)-(c)

(4) (i)-(c), (ii)-(b), (iii)-(a), (iv)-(d)

68. The oxidation electrode potentials of three metals A, B and C are –0.76 V, +0.8 V and –1.23 V respectively. The reducing power of these metals will be in the order of

(1) B > A > C

(2) A > B > C

(3) C > A > B

(4) B > C > A

69. Oxidation numbers of A, B and C are +3, +4 and –2 respectively. Possible formula of the compound is

(1) A_2BC_3

(2) A_2BC_5

(3) $A_2B_3C_5$

(4) AB_3C_5

70. For a redox reaction

$Cr_2O_7^{2-} + C_2O_4^{2-} + H^+ \rightarrow Cr^{3+} + CO_2 + H_2O$

The correct coefficient of the species for the balanced reaction are

	$Cr_2O_7^{2-}$	$C_2O_4^{2-}$	H_2O
(i)	2	6	5
(ii)	1	3	7
(iii)	1	6	5
(iv)	3	1	7

(1) (i)

(2) (ii)

(3) (iii)

(4) (iv)

71. Given below are two statements

Statement I : 0.5 molal aqueous solution of urea contains 30 g of urea in 1 kg water.

Statement II : Solution containing one mole of solute in 1000 g solvent is called one molal solution.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

72. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Aquatic species are more comfortable in cold water rather than in warm water.

Reason (R) : Solubility of gases increases with increase in temperature.

In the light of the above statements, choose the correct answer.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

73. Liquids 'A' and 'B' form an ideal solution. The vapour pressure of solution having 40 mole percent of A in the solution is

(P_A^0 = 80 cm Hg, P_B^0 = 30 cm Hg)

(1) 20 cm Hg

(2) 50 cm Hg

(3) 25 cm Hg

(4) 55 cm Hg

74. Consider the following statements

(i) Mixture of phenol and aniline shows negative deviation from Raoult's law.

(ii) Nitric acid and water form maximum boiling azeotrope.

(iii) Two solutions having same osmotic pressure at a given temperature are called isotonic solutions.

The correct statements(s) is/are

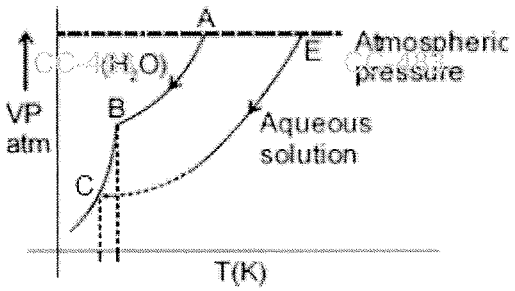
(1) (i) and (ii) only

(2) (ii) and (iii) only

(3) (i) only

(4) (i), (ii) and (iii)

75.



In the above graph which point indicates freezing point of H_2O?

(1) Point A

(2) Point B

(3) Point C

(4) Point E

76. 31 g of ethylene glycol ($C_2H_6O_2$) is mixed with 500 g of water. What is the freezing point of the solution? $[K_{f(H_2O)} = 1.86 Kkgmol^{-1}]$

(1) 271.14 K

(2) 275.01 K

(3) 273.15 K

(4) 269.43 K

77. 200 ml of an aqueous solution contains 18 g of glucose. The osmotic pressure of the solution at 27 °C will be (Molar mass of glucose = 180 g mol^{-1})

(1) 6.12 atm

(2) 8.25 atm

(3) 12.3 atm

(4) 10.5 atm

78. The limiting ionic molar conductivities of K^+, Al^{3+}, SO_4^{2-} ions are x, y and z S cm^2 mol^{-1} respectively. The λ_m^0 value for potash alum is

(1) (2x + y + 4z) S cm^2 mol^{-1}

(2) (2x + 2y + 4z) S cm^2 mol^{-1}

(3) (2x + 2y – 4z) S cm^2 mol^{-1}

(4) $\left(\frac{x}{4}+\frac{3y}{4}+z\right)$ S cm^2 mol^{-1}

79. Intercept of graph which is plotted between Λ_m and $c^{1/2}$ for strong electrolyte is (Λ_m plotted on y-axis)

(1) Λ_m^0

(2) $\sqrt{\kappa}$

(3) κ

(4) $\frac{1}{\Lambda_m^0}$

80. The weight of copper (At wt = 63.5) displaced by a quantity of electricity which displaces 11200 mL of O_2 gas at STP will be

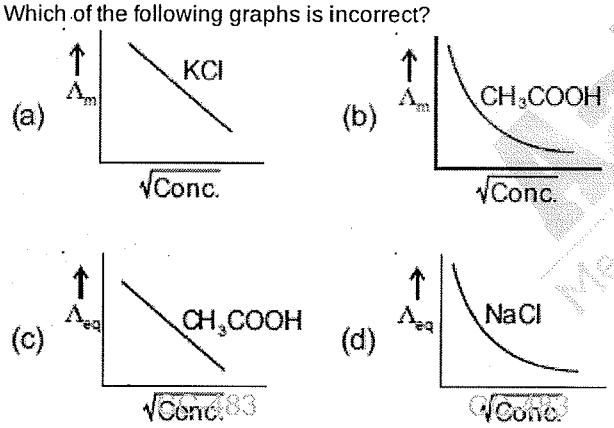
(1) 31.75 g

(2) 63.5 g

(3) 95.25 g

(4) 127 g

81. Which of the following graphs is incorrect?



(1) (a) & (b) only

(2) (a), (b) & (c) only

(3) (c) & (d) only

(4) (c) only

82. For the following cell reaction

$Ni(s) + Cu^{2+}(aq) \rightarrow Ni^{2+}(aq) + Cu(s)$

$E^{\circ}_{Ni^{2+}/Ni} = -0.25$ V and $E^{\circ}_{Cu^{2+}/Cu} = 0.34$ V. ΔG° for the reaction is

(1) –2.2 F

(2) –1.18 F

(3) –3.21 F

(4) –1.75 F

83. In mercury cell, the electrolyte used is

(1) A paste of NH_4Cl and $ZnCl_2$

(2) A paste of KOH and ZnO

(3) 38% solution of sulphuric acid

(4) 20% solution of NaCl

84. The value of ΔG° for the cell

Mg(s) | Mg^{2+}(0.01 M) || Cu^{2+} (0.01 M) | Cu(s) is

$\left(E^{\circ}_{Cu^{2+}/Cu} = 0.34\ V,\ E^{\circ}_{Mg^{2+}/Mg} = -2.37\ V\right)$

(1) –2.71 F J mol^{-1}

(2) 5.42 F J mol^{-1}

(3) –5.42 F J mol^{-1}

(4) 2.71 F J mol^{-1}

85. Anode and Cathode of lead storage battery respectively are made up of

(1) Lead plates, grid of Pb packed with Fe_2O_3

(2) Lead plates, grid of Pb packed with PbO_2

(3) Iron plates, grid of Pb packed with Fe_2O_3

(4) Iron plates, grid of Pb packed with PbO_2

SECTION-B

86. Consider the following reactions

Reaction	$\Delta H(kJ)$
$A \rightarrow B$	$+300$
$2B \rightarrow 3C + D$	-200
$E + C \rightarrow 4D$	$+700$

ΔH (in kJ) for the reaction 2A + 3E → 13D, will be

(1) –2000

(2) +1200

(3) +2500

(4) –2800

87. Entropy of vaporisation of water at 100°C, if molar heat of vaporisation is 7460 cal mol^{-1} will be

(1) 25 cal K^{-1} mol^{-1}

(2) 20 cal K^{-1} mol^{-1}

(3) 30 cal K^{-1} mol^{-1}

(4) 28 cal K^{-1} mol^{-1}

88. Given below are two statements one is assertion (A) other is reason (R).
Assertion (A) : The ionisation of CH_3COOH is suppressed on adding CH_3COONa.
Reason (R) : Common ion effect is observed on addition of acetate ions to acetic acid solution.
In the light of above statements choose the correct option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

89. With increase in temperature, pOH of pure water

(1) Increases

(2) Decreases

(3) May increase or decrease

(4) Remains constant

90. K_p and K'_p are the equilibrium constants of the two reaction, respectively given below

$\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \rightleftharpoons NH_3(g)$ K_p

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ K'_p

Therefore K_p and K'_p are related by

(1) $K_p = K'^2_p$

(2) $K_p = \sqrt{K'_p}$

(3) $K_p = 2K'_p$

(4) $K_p = K'_p$

91. In carbon suboxide, the oxidation state(s) of carbon are

(1) +2 and +3

(2) +2 and 0

(3) +2 only

(4) +1 and +3

92. How many moles of $Cr_2O_7^{2-}$ is needed to oxidise one mole of H_2S to sulphur in acidic medium?

(1) $\frac{1}{3}$

(2) 3

(3) $\frac{1}{2}$

(4) 2

93. Given below are two statements:
Statement-I: Oxidation number of Cr in CrO_5 is +6.
Statement-II: In CrO_5, five oxygen atoms are involved in peroxide linkage.
In the light of the above statements, choose the correct answer.

(1) Statement-I is correct but statement-II is incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Both statement-I and statement-II are correct

(4) Both statement-I and statement-II are incorrect

94. For $[CrCl_3.x\ NH_3]$, elevation in boiling point of one molal solution is double of one molal solution of glucose, hence x is , if complex is 100% ionised.

(1) 3

(2) 4

(3) 5

(4) 6

95. Solutions showing negative deviation from Raoult's law is

(1) $HCl + H_2O$

(2) $CCl_4 + CHCl_3$

(3) $CH_3OH + H_2O$

(4) $C_2H_5Br + C_2H_5I$

96. A is non-electrolyte, 1.02% (w/V) aqueous solution of A and 2% (w/V) aqueous solution of glucose are isotonic. Find the molecular mass of A

(1) 91.8

(2) 9.18

(3) 183.6

(4) 18.36

97. The molar conductance of $\frac{M}{64}$ solution of a weak monoacidic base is 5.0 S cm^2 mol^{-1} and at infinite dilution is 200 S cm^2 mol^{-1}. The concentration of OH^- in this solution is

(1) 1.25×10^{-5} M

(2) 3.90×10^{-4} M

(3) 6.25×10^{-4} M

(4) 1.25×10^{-4} M

98. Anode used in commercial dry cell is

(1) Graphite

(2) Zinc

(3) Copper

(4) Lead

99. What is the product of electrolysis of KNO_3(aq) using Pt electrodes?

(1) K, NO_2

(2) H_2, NO_2

(3) K, O_2

(4) H_2, O_2

100. The difference between heat of reaction at constant pressure and constant volume for the reaction $2C_6H_6(l) + 15O_2(g) \rightarrow 12CO_2(g) + 6H_2O(l)$ at 25°C in kJ is

(1) +7.43

(2) +4.83

(3) –7.43

(4) –4.83

BOTANY

SECTION-A

101. In leguminous plants, swollen leaf base is called

(1) Stipule

(2) Pulvinus

(3) Midrib

(4) Lamina

102. The arrangement of flowers on the floral axis is termed as

(1) Venation

(2) Floral symmetry

(3) Inflorescence

(4) Phyllotaxy

103. Seed coat has two layers, the outer A and the inner B. Select the **correct** option for A and B.

(1) **A**-Tegmen, **B**-Testa

(2) **A**-Testa, **B**-Tegmen

(3) **A**-Pericarp, **B**-Micropyle

(4) **A**-Hilum, **B**-Pericarp

104. Drupe fruit of mango

(1) Has fibrous mesocarp

(2) Develops from monocarpellary ovary

(3) Has stony epicarp

(4) Does not have well differentiated pericarp

105. Which of the following is an endospermous seed?

(1) Gram

(2) Castor

(3) Pea

(4) Bean

106. Stem

(1) Develops from radicle

(2) Bears nodes, internodes and buds

(3) Is involved in conduction of water only

(4) Does not perform the function of vegetative propagation

107. $\oplus \, ♂ \, K_{(5)} \, \overset{\frown}{C_{(5)} \, A_5} \, \underline{G}_{(2)}$

is the floral formula of a plant family whose one of the members is

(1) *Petunia*

(2) *Gloriosa*

(3) *Indigofera*

(4) *Trifolium*

108. Which of the given symbols is used in floral formula to represent epipetalous stamens?

(1) $\overset{\frown}{P\ A}$

(2) $\overset{\frown}{C\ A}$

(3) $C_{(n)}$

(4) $\overset{\frown}{K\ A}$

109. Choose the **mismatched** pair.

(1) Stem tendrils – Watermelon

(2) Stem thorns – *Citrus*

(3) Offsets – *Eichhornia*

(4) Cylindrical phylloclade – *Opuntia*

110. Read the following assertion and reason and choose the correct option.
Assertion : Outer covering of embryo in maize is called aleurone layer, separates the embryo from endosperm.
Reason : Endosperm in maize is small and situated in the groove of embryo.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

111. Identify the **wrongly** matched pair.

(1) Pinnately compound leaf – Neem

(2) Palmately compound leaf – Silk cotton

(3) Alternate phyllotaxy – China rose

(4) Opposite phyllotaxy – Mustard

(1) (1)

(2) (2)

(3) (3)

(4) (4)

112. Match the columns w.r.t. placentation.

	Column I		Column II
a.	Axile	(i)	Pea plant
b.	Basal	(ii)	*Argemone*
c.	Parietal	(iii)	Sunflower
d.	Marginal	(iv)	Lemon

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iv), b(ii), c(i), d(iii)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(iv), b(iii), c(i), d(ii)

113. Mesophyll cells are divided into palisade and spongy parenchyma in

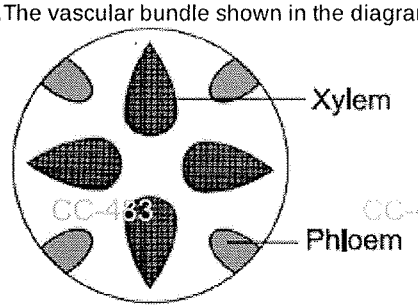
(1) Grass leaf

(2) Monocot leaf

(3) Isobilateral leaf

(4) Dicot leaf

114. The vascular bundle shown in the diagram is present in



(1) Dicot leaf

(2) Monocot stem

(3) Dicot root

(4) Dicot stem

115. All are true for trichomes, **except**

(1) They may be branched or unbranched

(2) They may be soft or stiff

(3) They help in transpiration as well as respiration

(4) They are epidermal hair of stem

116. Companion cells are specialised

(1) Parenchymatous cells

(2) Sclerenchymatous cells

(3) Collenchymatous cells

(4) Enucleated cells

117. In grasses, guard cells of stomata are

(1) Bean-shaped

(2) Kidney-shaped

(3) Sickle shaped

(4) Dumb-bell shaped

118. Read the following statements and select the **correct** option.
Statement A: Collenchyma forms the major component within a plant organ.
Statement B: Xylem fibres have highly thickened lignified walls with obliterated central lumen.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are correct

(4) Both statements are incorrect

119. Match the following columns and select the **correct** option.

	Column I		Column II
(a)	Sieve tube elements	(i)	Store resins, tannins, mucilage, latex etc.
(b)	Companion cells	(ii)	Help in maintaining pressure gradient
(c)	Phloem fibres	(iii)	Possess a peripheral cytoplasm but lack a nucleus
(d)	Phloem parenchyma	(iv)	Have pointed, needle- like apices

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(ii), b(i), c(iii), d(iv)

120. Which of the following is **not** a part of stele?

(1) Pith

(2) Xylem

(3) Phloem

(4) Cortex

121. Tissue is a group of cells which
(a) Usually perform a common function.
(b) Are structurally always similar.
(c) Have a common origin.

(1) Only (a) and (b) are correct

(2) Only (a) and (c) are correct

(3) All (a), (b) and (c) are correct

(4) All are correct except (a)

122. Select the **odd** one w.r.t. ground tissue system.

(1) Epidermis

(2) Pith

(3) Pericycle

(4) Cortex

123. Intercalary meristems are intercalated between the

(1) Permanent tissues

(2) Cells of meristematic tissues

(3) Cells of lateral meristem

(4) Cells of root cap

124. Collenchyma differs from sclerenchyma as the former

(1) Is a living tissue

(2) Is a mechanical tissue

(3) Has lignin deposition at the corners of cell wall

(4) Loses protoplasm at maturity

125. Bulliform cells are

(1) Green coloured photosynthetic cells in dicots

(2) Modified adaxial epidermal cells in grasses

(3) Helpful to maximise water loss in monocots

(4) Small endodermal cells in maize

126. Starch sheath is present in

(1) Dicot root

(2) Dicot stem

(3) Monocot root

(4) Monocot stem

127. Choose the **odd** one w.r.t. monocot stem.

(1) Sclerenchymatous hypodermis

(2) Scattered vascular bundles

(3) Sclerenchymatous bundle sheath

(4) Larger peripheral vascular bundles

128. Antherozoids of bryophytes are

(1) Biflagellated

(2) Uniflagellated

(3) Multiciliated

(4) Non-motile

129. In pteridophytes, leaves bearing sporangia are called

(1) Strobili

(2) Sporophyte

(3) Cones

(4) Sporophylls

130. Pyrenoids in green algae contain

(1) Protein and starch

(2) Cellulose and fatty acids

(3) Cellulose and starch

(4) Glucose and glycogen

131. Pteridophytes have all, **except**

(1) True leaves

(2) Zygote

(3) Embryo

(4) Seed

132. Which of the given statement is **incorrect** w.r.t. 'numerical taxonomy'?

(1) Involves usage of numerical methods.

(2) Numbers and codes are assigned to all observable characters.

(3) Is based on only chemical constituent of plant.

(4) Organisation and analysis of data forms core of numerical taxonomy.

133. Match the following columns and select the **correct** option

	Column-I (Classes)		Column-II (Examples)
a.	Psilopsida	(i)	*Selaginella*
b.	Lycopsida	(ii)	*Psilotum*
c.	Sphenopsida	(iii)	*Dryopteris*
d.	Pteropsida	(iv)	*Equisetum*

(1) a(ii), b(i), c(iv), d(iii)

(2) a(i), b(ii), c(iv), d(iii)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(ii), b(iv), c(i), d(iii)

134. Protonema found in mosses is
(a) Haploid in nature.
(b) A part of capsule.
(c) Vascular and non-green.
(d) Non-green and branched.

(1) Only (a)

(2) Only (b) & (c)

(3) All (a), (b), (c) & (d)

(4) Only (a) & (c)

135. Colonial alga is

(1) *Volvox*

(2) *Spirogyra*

(3) *Ulothrix*

(4) *Nostoc*

SECTION-B

136. Superior ovary is found in

(1) Hypogynous flower

(2) Perigynous flower

(3) Epigynous flower

(4) Ray florets of sunflower

137. Keel is the characteristic feature of the flowers of

(1) China rose

(2) Soyabean

(3) Tulip

(4) Tobacco

138. In cymose inflorescence
(a) Main axis terminates into a flower.
(b) Flowers are borne in basipetal order.
(c) Younger flowers are present towards the apex and older at the base.

(1) Only (c) is correct

(2) All (a), (b) and (c) are correct

(3) Only (a) and (b) are correct

(4) Only (a) is correct

139. Primary root is short lived and a large number of roots originate from the base of stem constituting A as found in B.

Fill in the blanks with **correct** option for A and B.

(1) A - Tap root system, B - Dicots

(2) A - Adventitious root system, B- Dicots

(3) A - Fibrous root system, B- Monocots

(4) A - Fibrous root system, B - Dicots

140. Tissue found in pulp of guava and pear which provide them gritty texture, is

(1) Sclereids

(2) Collenchyma

(3) Parenchyma

(4) Sclerenchyma fibres

141. Lateral roots arise from

(1) Epidermis

(2) Pericycle

(3) Hypodermis

(4) Pith

142. An axillary bud

(1) Is a secondary meristem

(2) Is found in the axil of leaf

(3) Is formed by root apical meristem

(4) Always form a flower

143. Consider the following statements and state them as **true (T)** or **false (F)**.
a. Epidermis is usually single layered
b. Cuticle is absent in roots.
c. Stomata cannot regulate the process of transpiration but can regulate gaseous exchange.
d. The cells of root epidermis bear a number of hairs.

(1) a(T), b(F), c(F), d(T)

(2) a(T), b(T), c(F), d(T)

(3) a(F), b(F), c(T), d(F)

(4) a(T), b(T), c(F), d(F)

144. Water containing cavities are seen inside the vascular bundles of

(1) Dicot root

(2) Monocot stem

(3) Dicot stem

(4) Monocot root

145. Conjunctive tissue lies between

(1) Epidermis and cortex

(2) Endodermis and pericycle

(3) Xylem and phloem

(4) Cortex and endodermis

146. Match the following columns and select the **correct** options.

a. *Salvinia*	(i)	Produce two kinds of spores
b. *Marchantia*	(ii)	Leaves are needle-like
c. *Pinus*	(iii)	Forms photosynthetic asexual buds
d. *Cycas*	(iv)	Roots associated with nitrogen fixing cyanobacteria

(1) a(ii), b(i), c(iv), d(iii)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(i), b(ii), c(iv), d(iii)

(4) a(iii), b(i), c(ii), d(iv)

147. Select the **incorrect** statement w.r.t. bryophytes.

(1) Commonly growing in moist and shaded areas

(2) Dependent on water for sexual reproduction

(3) Lack true roots, stem and leaves

(4) Zygote undergoes reduction division to form spore

148. Leaves of conifers do **not** have

(1) Needle like shape

(2) Very thin cuticle

(3) Sunken stomata

(4) Ability to withstand extremes of temperature

149. Pyriform gametes with laterally attached flagella are present in

(1) Red algae

(2) Brown algae

(3) Green algae

(4) Blue green algae

150. Gemmae are

(1) Green, multicellular and asexual buds

(2) Green, unicellular and asexual buds

(3) Non-green, unicellular and sexual spores

(4) Non-green, multicellular and sexual spores

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ZOOLOGY

SECTION-A

151. Generation of second messengers is associated with the mechanism of action of which of the following set of hormones?

(1) Estrogen and ACTH

(2) Insulin and TSH

(3) Cortisol and ACTH

(4) Progesterone and androgens

152. How many bones given in the box below are not included in cranium?

Ethmoid, Temporal, Zygomatic, Maxilla, Frontal, Vomer, Lacrimal

(1) Three

(2) Four

(3) Five

(4) Six

153. The hormone which stimulates secretion of water and bicarbonate ions from exocrine pancreas in humans is

(1) Gastrin

(2) Secretin

(3) Cholecystokinin

(4) GIP

154. Sweat produced by the sweat glands is a watery fluid which does **not** contain

(1) NaCl

(2) Urea

(3) Lactic acid

(4) Sterols

155. When a neuron is not conducting any impulse *i.e.* it is at resting stage, the axonal membrane show all of the following events, **except**

(1) Permeability for K^+

(2) Nearly impermeable for Na^+

(3) Impermeability for negatively charged proteins

(4) Active transport of $2Na^+$ outside for $3K^+$ into the cell

156. **Assertion (A) :** Initial signal for micturition is provided by the stretching of urinary bladder.
Reason (R) : Relaxation of urethral sphincter causes micturition.
In the light of above statements choose the **correct** answer from the options given below.

(1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

157. Which segment allows passage of small amounts of urea into the medullary interstitium to maintain the osmolarity?

(1) PCT

(2) DCT

(3) Collecting duct

(4) Henle's Loop

158. Order of toxicity among nitrogenous waste products from higher to lower is

(1) Uric acid > ammonia > urea

(2) Uric acid < ammonia < urea

(3) Urea > uric acid > ammonia

(4) Ammonia > urea > uric acid

159. Brain stem forms the connection between the brain and spinal cord. The major regions that make up brain stem include all, **except**

(1) Pons

(2) Medulla oblongata

(3) Cerebrum

(4) Midbrain

160. The somatic neural system relays impulses from CNS to muscles attached to/present in __A__ while the autonomic neural system transmits impulses from the CNS to the muscles found in __B__.
Select the option that correctly identifies 'A' and 'B' respectively

(1) Bone, Heart

(2) Heart, Iris of eyeball

(3) Stomach, Oesophagus

(4) Heart, Small intestine

161. Hypersecretion of growth hormone since childhood causes

(1) Pituitary dwarfism

(2) Addison's disease

(3) Gigantism

(4) Acromegaly

162. Which of the following hormones acts on the mammary glands and stimulates the formation of sac-like structures which store milk in them?

(1) Progesterone

(2) Oxytocin

(3) Prolactin

(4) Luteinizing hormone

163. Which of the following features differentiates open circulatory system from closed type of circulatory system?

(1) Presence of heart

(2) Presence of arteries

(3) Absence of capillaries

(4) Absence of circulatory fluid

164. Select the option which represents correct sequence w.r.t cross bridge formation and its breakdown.
(a) Binding of new ATP with myosin head
(b) Cross bridge breaks
(c) Hydrolysis of ATP by the ATPase activity
(d) Cross bridge formation
(e) Sliding of actin filament

(1) c → d → e → a → b

(2) b → a → d → c → e

(3) e → a → b → d → c

(4) c → a → d → b → e

165. An auto-immune disorder affecting neuro-muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is

(1) Myasthenia gravis

(2) Muscular dystrophy

(3) Tetany

(4) Arthritis

166. Arrange the following vertebrae in sequential order starting from neck.

(1) Cervical – Lumbar – Thoracic – Sacral – Coccygeal

(2) Cervical – Thoracic – Sacral – Lumbar – Coccygeal

(3) Cervical – Sacral – Thoracic – Lumbar – Coccygeal

(4) Cervical – Thoracic – Lumbar – Sacral – Coccygeal

167. **Assertion (A)** : Removal of PCT results in formation of large amounts of dilute urine.
Reason (R) : 70–80 percent of electrolytes and water are reabsorbed by DCT.
In the light of above statements, choose the **correct** option.

(1) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of the Assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

168. 'X' and 'Y' are hormones, 'X' stimulates the secretion of 'Y' which exerts negative feedback on the cells that secrete 'X'. What happens when the blood level of 'Y' decreases?

(1) Less 'X' is secreted

(2) More 'X' is secreted

(3) Secretion of 'X' stops

(4) Blood levels of 'X' and 'Y' will continuously decrease

169. Which of the following is **incorrect** w.r.t. ANF?

(1) Causes vasodilation

(2) Keeps a check on the renin-angiotensin mechanism

(3) Increases blood pressure

(4) Causes natriuresis/diuresis

170. Read the given statements.
Statement I : Each coxal bone in humans is formed by the fusion of three bones namely ilium, ischium and pubis.
Statement II : The two halves of the pelvic girdle meet dorsally to form the pubic symphysis containing elastic cartilage.
Select the **correct** option.

(1) Only statement II is correct

(2) Only statement I is correct

(3) Both statements I and II are correct

(4) Both statements I and II are incorrect

171. All of the following hormones increase formation of red blood cells in our body, **except**

(1) Erythropoietin

(2) Cortisol

(3) Thyroxine

(4) Oestrogen

172. Which of the following diseases is characterized by stunted growth, mental retardation, low intelligence quotient, abnormal skin and deaf-mutism?

(1) Hyperthyroidism

(2) Cretinism

(3) Pituitary dwarfism

(4) Goitre

173. A part of brain that lies at the base of thalamus and controls body temperature, urge for eating and drinking is

(1) Amygdala

(2) Hippocampus

(3) Hypothalamus

(4) Cerebrum

174. Select the **correct** pair where Nissl's granules are found

(1) Axon and dendrites

(2) Cell body and dendrites

(3) Cyton and axon

(4) Cyton and nucleus

175. Four rounded swellings called 'corpora quadrigemina' constitute mainly

(1) Floor of thalamus

(2) Cerebellum

(3) Dorsal portion of midbrain

(4) Pons varolii

176. 'The former is a sessile and cylindrical form whereas the latter is umbrella-shaped and free-swimming form of coelenterates.' The former and the latter respectively, are

(1) Polyp; Medusa

(2) Medusa; polyp

(3) *Aurelia*; *Adamsia*

(4) Sexually reproducing stage; Asexually reproducing stage

177. Alimentary canal is complete with a well developed muscular pharynx in all of these animals, **except**

(1) Hookworm

(2) Tapeworm

(3) Round worm

(4) Filarial worm

178. Choose the correct option to complete the analogy.
Collar cells : Bath sponge : : Cnidocytes :

(1) *Spongilla*

(2) *Euspongia*

(3) *Meandrina*

(4) *Sycon*

179. Read the given statements and choose the correct option.
Statement - I : Nematodes exhibit sexual dimorphism.
Statement - II: In nematodes, males are longer and larger than females.

(1) Both statements I and II are correct

(2) Both statements I and II are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

180. Choose the animal which has no brain but have nerve cells

(1) *Hydra*

(2) *Pheretima*

(3) *Periplaneta*

(4) *Euspongia*

181. In *Pleurobrachia*

(1) Fertilization is external with direct development

(2) Fertilization is internal with direct development

(3) Fertilization is external with indirect development

(4) Fertilization is internal with indirect development

182. Limbic system along with hypothalamus is involved in regulation of all, **except**

(1) Sexual behaviour

(2) Expression of emotional reactions

(3) Motivation

(4) Cardiovascular reflexes

183. Neuron with one axon and only one dendrite is called

(1) Multipolar

(2) Bipolar

(3) Unipolar

(4) Apolar

184. JGA is a specialized sensitive region formed by the cellular modification in the

(1) PCT and efferent arteriole

(2) DCT and afferent arteriole

(3) DCT and efferent arteriole

(4) PCT and glomerulus

185. Cell aggregate body plan is exhibited by

(1) Sponges

(2) Cnidarians

(3) Flatworms

(4) Roundworms

SECTION-B

186. In an adult human, vertebrae of which category are fused?

(1) Sacral and coccygeal

(2) Cervical and thoracic

(3) Thoracic and lumbar

(4) Cervical and coccygeal

187. Juxtamedullary nephrons differ from cortical nephrons in all of the following, **except**

(1) Number

(2) Length of loop of Henle

(3) Presence of vasa recta which runs deep into medulla

(4) Presence of peritubular capillary network

188. Choose the correct option to complete the analogy for the following
Flagellar movement : Spermatozoa : : Amoeboid movement : _____

(1) Ciliated epithelial cells

(2) RBCs

(3) Leucocytes

(4) Cuboidal epithelial cells

189. Glucocorticoids promote all of the following, **except**

(1) Gluconeogenesis

(2) Lipolysis

(3) Proteolysis

(4) Storage of glucose in liver

190. Association areas in the forebrain of humans are responsible for all of the following functions, **except**

(1) Intersensory association

(2) Regulation of gastric secretions

(3) Memory

(4) Communication

191. Gonadotropin releasing hormone and gonadotrophins are respectively the products of

(1) Posterior pituitary and hypothalamus

(2) Hypothalamus and posterior pituitary

(3) Hypothalamus and anterior pituitary

(4) Anterior pituitary and hypothalamus

192. Acromion process articulates with the

(1) Head of femur

(2) Head of humerus

(3) Collar bone

(4) Sternum

193. Read the following statements and choose the correct option.
Statement A: Red muscle fibres have few mitochondria in them, but exhibit high amount of sarcoplasmic reticulum.
Statement B: White muscle fibres are also termed as aerobic muscle fibres as they produce huge amount of ATP.

(1) Both statements A and B are incorrect

(2) Both statements A and B are correct

(3) Only statement A is correct

(4) Only statement B is correct

194. Hormone synthesised by adenohypophysis is

(1) Oxytocin

(2) Vasopressin

(3) Renin

(4) Prolactin

195. _____ hormone causes dilation of blood vessels, increased oxygen consumption and glycogenolysis in skeletal muscles.
Choose the option which fills the blank **correctly**.

(1) Adrenaline

(2) Glucagon

(3) ACTH

(4) Insulin

196. How many among the following given organisms exhibit bilateral symmetry?
a. Hookworm
b. Liver fluke
c. Roundworm
d. Filarial worm
Choose the **correct** option

(1) Two

(2) One

(3) Three

(4) Four

197. Select the **incorrect** statement w.r.t. transmission of an impulse across synapses.

(1) Impulse transmission across an electrical synapse is always faster than that across a chemical synapse.

(2) Chemical synapses require neurotransmitters for transmission of impulses.

(3) Fluid filled synaptic cleft is always present in both chemical and electrical synapses.

(4) Electrical synapses are rare in our body system.

198. In a neuron, the vesicles containing neurotransmitters are located in the

(1) Nucleus

(2) Synaptic cleft

(3) Synaptic knob

(4) Cell body

199. The part of renal tubule which is impermeable to water is

(1) Descending limb of loop of Henle

(2) Ascending limb of loop of Henle

(3) PCT

(4) Collecting duct

200. Select the set of ureotelic animals.

(1) Bony fishes, aquatic amphibians, aquatic insects

(2) Marine fishes, terrestrial amphibians, mammals

(3) Mammals, land snails, insects

(4) Reptiles, birds and mammals

13-11-2024

Code-B

Phase-B_RM

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NCERT BOOSTER TEST SERIES

(*for* NEET-2025)

Test - 2

Answers

1. (1)	41. (4)	81. (3)	121. (2)	161. (3)
2. (2)	42. (1)	82. (2)	122. (1)	162. (1)
3. (1)	43. (1)	83. (2)	123. (1)	163. (3)
4. (1)	44. (1)	84. (3)	124. (1)	164. (1)
5. (3)	45. (4)	85. (2)	125. (2)	165. (1)
6. (1)	46. (3)	86. (3)	126. (2)	166. (4)
7. (2)	47. (2)	87. (2)	127. (4)	167. (3)
8. (3)	48. (2)	88. (1)	128. (1)	168. (2)
9. (1)	49. (3)	89. (2)	129. (4)	169. (3)
10. (3)	50. (3)	90. (2)	130. (1)	170. (2)
11. (2)	51. (1)	91. (2)	131. (4)	171. (4)
12. (2)	52. (4)	92. (1)	132. (3)	172. (2)
13. (2)	53. (2)	93. (1)	133. (1)	173. (3)
14. (2)	54. (2)	94. (2)	134. (1)	174. (2)
15. (3)	55. (2)	95. (1)	135. (1)	175. (3)
16. (1)	56. (2)	96. (1)	136. (1)	176. (1)
17. (4)	57. (3)	97. (2)	137. (2)	177. (2)
18. (3)	58. (1)	98. (2)	138. (3)	178. (3)
19. (4)	59. (2)	99. (4)	139. (3)	179. (3)
20. (2)	60. (3)	100. (3)	140. (1)	180. (1)
21. (4)	61. (2)	101. (2)	141. (2)	181. (3)
22. (2)	62. (2)	102. (3)	142. (2)	182. (4)
23. (4)	63. (1)	103. (2)	143. (2)	183. (2)
24. (3)	64. (1)	104. (2)	144. (2)	184. (2)
25. (1)	65. (4)	105. (2)	145. (3)	185. (1)
26. (4)	66. (3)	106. (2)	146. (2)	186. (1)
27. (2)	67. (2)	107. (1)	147. (4)	187. (4)
28. (2)	68. (1)	108. (2)	148. (2)	188. (3)
29. (4)	69. (2)	109. (4)	149. (2)	189. (4)
30. (2)	70. (2)	110. (4)	150. (1)	190. (2)
31. (3)	71. (1)	111. (4)	151. (2)	191. (3)
32. (2)	72. (3)	112. (1)	152. (2)	192. (3)
33. (2)	73. (2)	113. (4)	153. (2)	193. (1)
34. (4)	74. (4)	114. (3)	154. (4)	194. (4)
35. (2)	75. (2)	115. (3)	155. (4)	195. (1)
36. (1)	76. (1)	116. (1)	156. (2)	196. (4)
37. (1)	77. (3)	117. (4)	157. (3)	197. (3)
38. (1)	78. (2)	118. (2)	158. (4)	198. (3)
39. (3)	79. (1)	119. (3)	159. (3)	199. (2)
40. (2)	80. (2)	120. (4)	160. (1)	200. (2)